

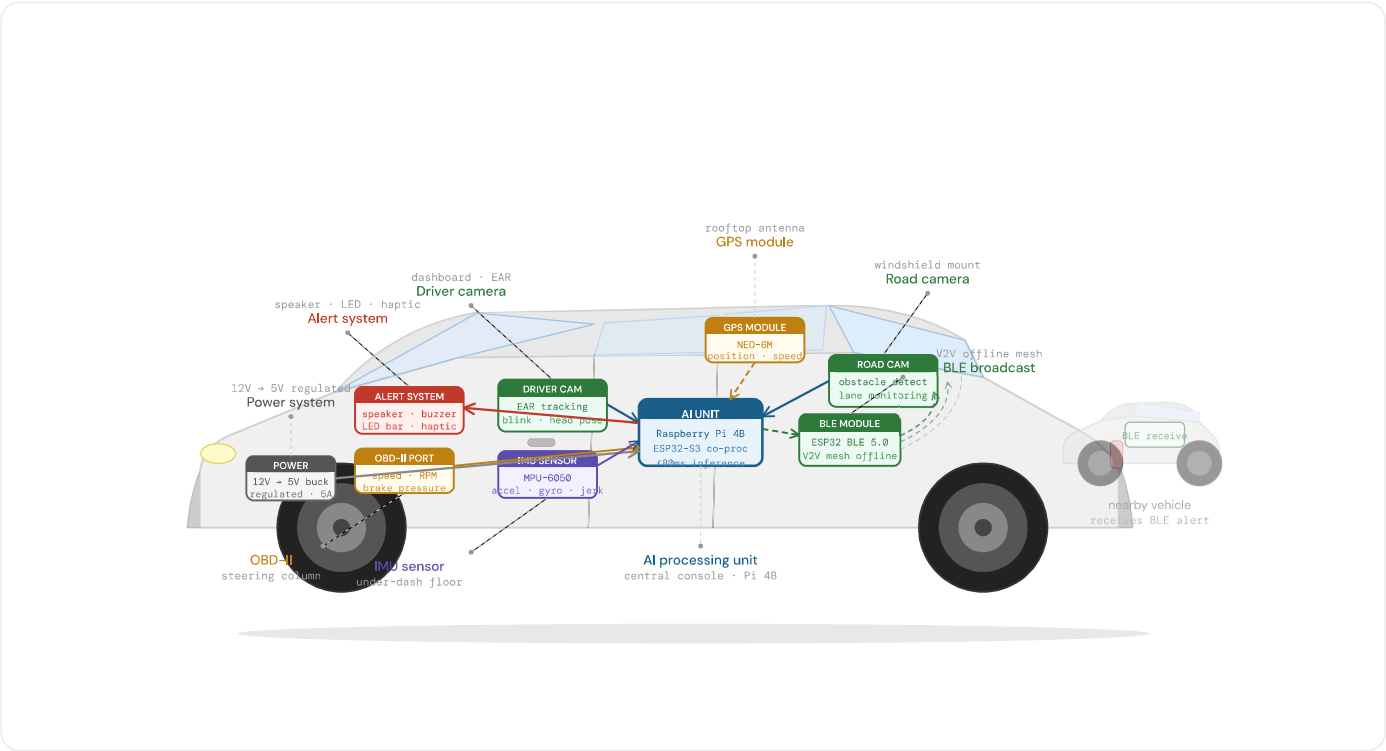
IN-VEHICLE INSTALLATION DIAGRAM

Project Garuda

AI-powered intelligent co-driver · accident prevention system · component layout & wiring

INFERENCE	BLE RANGE	POWER
< 80 ms	~100 m	12 V → 5 V

→ Chrome: Ctrl + Shift + P → "Capture full size screenshot" to save as image



▶ Data flow Sensor → AI unit	Power / sensor wire 12V battery · GPS · OBD-II	Alert output ▶ AI → speaker · LED · buzzer	BLE wireless ----- Offline V2V · no internet	Power wiring ▶ Battery → buck → components
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CORE AI AI processing unit Raspberry Pi 4B · 4GB · ESP32-S3 co-proc · TFLite inference <80ms	VISION Camera system Driver cam (EAR) · Road cam (obstacle detect) · 20–30 fps	SENSORS GPS + OBD-II NEO-6M position · ELM327 speed · RPM · brake pressure	MOTION IMU sensor MPU-6050 · acceleration · gyroscope · jerk · yaw rate
OUTPUT Alert system Speaker + TTS · buzzer · RGB LED bar · <100ms trigger	COMMS BLE V2V module BLE 5.0 · offline mesh · <200ms broadcast · ~100m range	POWER Power system 12V car battery → 5V/5A regulated · TVS spike protection	CLOUD 4G / Wi-Fi sync Optional cloud OTA · model updates · emergency SOS broadcast